

Week 1: Introduction to the course

Physics classes, M1 introductory module
Master in Molecular Nanoscience and Nanotechnology
ICMol Seminar room 0.10.6

Topics:

Getting to know each other (class self-awareness). Goal of the course and teaching philosophy. Limitations of the course. Mechanics of the course. Introduction to the student journal club.

Timetable:

28th of September

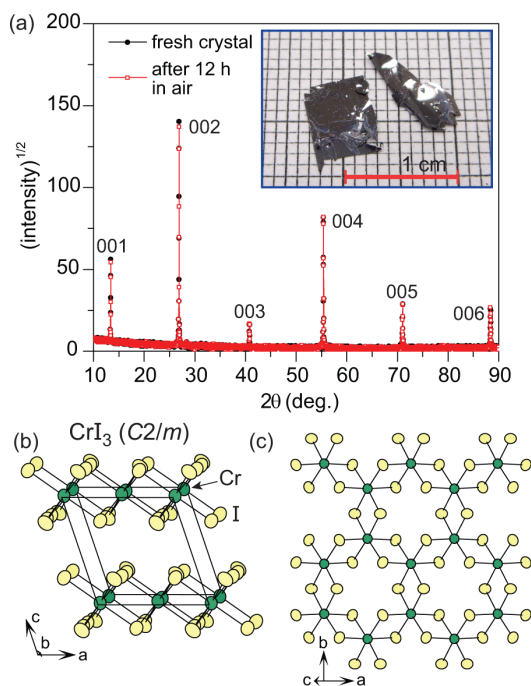
15.00–15.10	Know your fellow master students exercise! Whole class
15.10–15.30	Goal, limitations and mechanics of the course. <i>Reading together the first handout.</i> Efrén Navarro
15.30–16.00	Introduction to the journal club and slot allocation. Efrén Navarro
16.00–16.30	Coupling of Crystal Structure and Magnetism in the Layered, Ferromagnetic Insulator CrI ₃ M. A. McGuire et al. <i>Chem. Mater.</i> 27 (2), 612–620 (2015). Efrén Navarro Full paper: page 6

Journal club handouts

Coupling of crystal structure and magnetism in the layered, ferromagnetic insulator CrI₃

28th of September 2026- Student Condensed Matter Physics journal club meeting (Parc Científic, Paterna)

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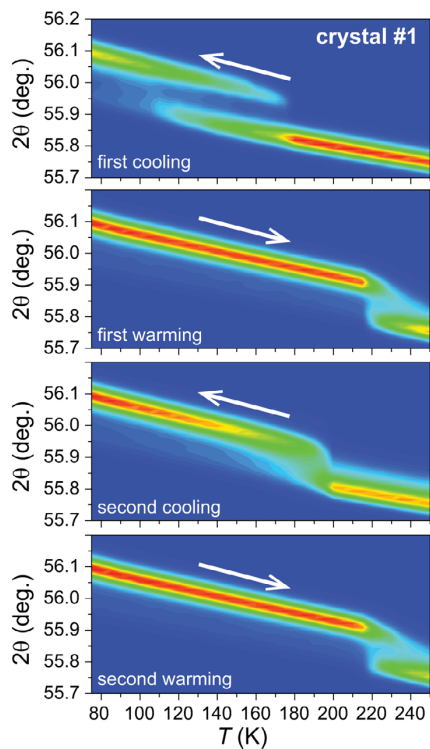
3

T (K)	250	90
space group	$C2/m$	$R\bar{3}^a$
a (Å)	6.866(3)	6.867(4)
b (Å)	11.886(5)	6.867(4)
c (Å)	6.984(3)	19.807(17)
β (deg)	108.507(7)	90
vol (Å ³)	540.5(4)	809(1)
Cr1 position	0, 0.16654(5), 0	$1/3, 2/3, 0.33299(7)$
I1 position	0.22407(5), 0, 0.23862(4)	0.31677(9), 0.33453(5), 0.41230(3)
I2 position	0.25110(3), 0.32452(2), 0.23467(4)	
R1/wR2, all data	0.020/0.049	0.057 [0.031] ^b /0.123 [0.069] ^b
Cr–I distances (Å)	2.7267(8), 2.7185(8), 2.7196(8)	2.725(2), 2.727(2)
Cr–Cr distances (Å)	3.959(2), 3.965(1)	3.965(2), 3.965(3)
I–I intralayer short distances (Å)	3.852(1), 3.862(1), 3.863(1)	3.857(2), 3.878(2)
I–I intralayer long distances (Å)	4.172(2), 4.184(1)	4.168(3)
I–I interlayer avg distances (Å)	4.155(1)	4.160(3)
layer spacing (Å)	6.623	6.602
vdW gap (Å)	3.488	3.471

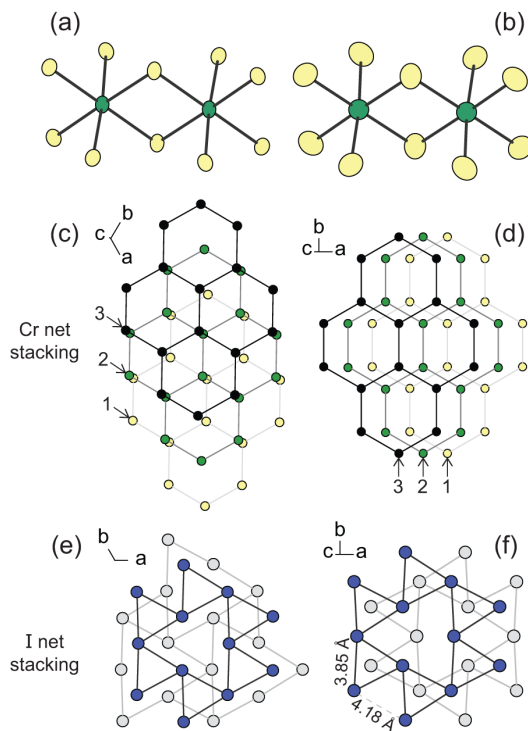
^aReverse-obverse twinning present with a twin fraction of 0.60.

^bValues in square brackets are for the model including a small amount of disorder described in the text and the SI.

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4



M. McGuire *et al.* *Chem. Mater.* **27**, 612-620 (2015)

Full papers

Topic: Student journal club demo

Time: 16.00–16.30

Coupling of Crystal Structure and Magnetism in the Layered, Ferromagnetic Insulator CrI_3

M. A. McGuire et al. *Chem. Mater.* **27**(2), 612–620 (2015).

Efrén Navarro

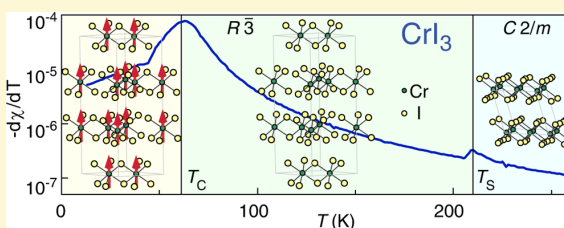
Coupling of Crystal Structure and Magnetism in the Layered, Ferromagnetic Insulator CrI₃

Michael A. McGuire,* Hemant Dixit, Valentino R. Cooper, and Brian C. Sales

Materials Science and Technology Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee United States

S Supporting Information

ABSTRACT: We have examined the crystallographic and magnetic properties of single crystals of CrI₃, an easily cleavable, layered and insulating ferromagnet with a Curie temperature of 61 K. Our X-ray diffraction studies reveal a first-order crystallographic phase transition occurring near 210–220 K upon warming, with significant thermal hysteresis. The low-temperature structure is rhombohedral ($R\bar{3}$, BiI₃-type) and the high-temperature structure is monoclinic ($C2/m$, AlCl₃-type). We find evidence for coupling between the crystallographic and magnetic degrees of freedom in CrI₃, observing an anomaly in the interlayer spacing at the Curie temperature and an anomaly in the magnetic susceptibility at the structural transition. First-principles calculations reveal the importance of proper treatment of the long-ranged interlayer forces, and van der Waals density functional theory does an excellent job of predicting the crystal structures and their relative stability. Calculations also suggest that the ferromagnetic order found in the bulk material may persist into monolayer form, suggesting that CrI₃ and other chromium trihalides may be promising materials for spintronic and magnetoelectronic research.



INTRODUCTION

There is tremendous interest in the development of two-dimensional (2D) materials as a means to accelerate size reduction and efficiency improvements in postsilicon microelectronics.^{1,2} It is also desirable to combine other functionalities in such electronic materials as a means to increase the versatility or performance of devices made from them. For example, spintronic devices, which use electron spins for information storage and processing, can be realized when magnetism is incorporated into the active materials.^{3,4} While graphene has been the focus of the majority of 2D electronic materials research to date, semiconducting transition-metal dichalcogenides like MoS₂ have more recently been identified as promising candidate materials, and devices based on them have been demonstrated using single- or few-layered dichalcogenides.^{5–7} One intrinsic advantage of the layered transition-metal dichalcogenides over graphene is that some of them, like MoS₂, have electronic band gaps, facilitating their application in devices like transistors. Chromium trihalides, CrX₃ (X = Cl, Br, I), are another class of van der Waals (vdW) bonded, layered semiconductors.^{8–11} In addition, CrX₃ compounds are magnetic, similar to the recently studied layered ternary compounds CrSiTe₃ and CrGeTe₃,¹² which is important for spintronic and magnetoelectronic applications.³ Chromium trihalides have been known for many decades; however, they have received relatively little attention, especially of late, and are not particularly well understood.

In chromium trihalides, Cr³⁺ ions are arranged in a honeycomb network in edge-sharing octahedral coordination by six X[−] ions, which are each bonded to two Cr ions. The

resulting slabs of composition CrX₃ are stacked with vdW gaps separating them. The structure type is then determined by the relative displacement of neighboring slabs and the stacking sequence. CrCl₃ adopts the monoclinic AlCl₃ structure type (space group $C2/m$) at room temperature and the rhombohedral BiI₃ structure type (space group $R\bar{3}$) below about 240 K.⁹ The same crystallographic phase transition occurs in CrBr₃ but at about 420 K,⁹ so that the room temperature structure for the tribromide is rhombohedral.^{8,13} Both CrCl₃ and CrBr₃ are Mott–Hubbard insulators.¹⁰ The chloride orders antiferromagnetically below about 17 K, with Cr magnetic moments within each CrCl₃ layer aligned ferromagnetically and the layers stacked antiferromagnetically. The moments lie within the layers but can be rotated with applied magnetic fields of only a few kilooersted to align ferromagnetically along the direction of the external field.^{14,15} CrBr₃ is ferromagnetic below its Curie temperature (T_C) of 37 K, and unlike the chloride, the bromide has Cr moments directed normal to the CrX₃ layers and more magnetic anisotropy is observed in the magnetically ordered state.¹⁶

The present work focuses on CrI₃. Among the chromium trihalides, it has the highest reported magnetic ordering temperature ($T_C = 68$ K)¹⁷ and the strongest magnetic anisotropy,¹⁸ and it is the simplest to prepare because iodine is solid at room temperature and can be handled relatively easily. It has a measured band gap near 1.2 eV.¹⁷ In addition, CrI₃ is

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cleaved very easily and is stable in air, and even in water, in the absence of CrI_2 contamination.¹⁹ Despite these attractive properties, this material is the least well understood among the CrX_3 compounds.

Surprisingly, the crystal structure of CrI_3 has not been clearly identified. The original powder diffraction study of CrI_3 assigned the noncentrosymmetric space group $P3_112$.⁸ About 12 years later, Morosin and Narath speculated that this may be incorrect because both the chloride and bromide had been found to adopt monoclinic and rhombohedral structures.⁹ However, no crystallographic studies have been reported to support this or to determine which of the two structure types might be stable at room temperature: the monoclinic AlCl_3 type found in CrCl_3 or the rhombohedral BiI_3 type adopted by CrBr_3 .

The magnetic properties of CrI_3 also have not been reported in any detail. Available magnetization data are limited to the Curie temperature and saturation moment at low temperature,¹⁷ magnetization versus field data at $T = 4.2$ K limited to fields too low to saturate the magnetic moment,¹⁸ and magnetic susceptibility data over a limited temperature range that does not show the ferromagnetic ordering transition.²⁰ The direction of the magnetic moment, perpendicular to the CrI_3 planes, has been inferred from Mössbauer spectroscopy.^{21,22}

Here we report the findings from our crystallographic and magnetic studies of CrI_3 single crystals grown by chemical vapor transport. We find that this compound undergoes a phase transition from the monoclinic AlCl_3 type to the rhombohedral BiI_3 type upon cooling from room temperature. We observe thermal hysteresis in the measured structural transition temperature (T_S) and a range of temperatures over which the two structure types coexist, consistent with a first-order crystallographic phase transition. Using single-crystal X-ray diffraction, we have refined fully the crystal structure in both the high- and low-temperature phases and identify both reverse-obverse twinning and chemical disorder in the rhombohedral phase, expected to arise from stacking faults developed at the phase transition. Our magnetization and heat capacity measurements reveal that long-range ferromagnetic order develops below $T_C = 61$ K, and there is evidence for strong magnetic anisotropy developing upon cooling below T_C . Through careful examination of our crystallographic and magnetic data, we find both a structural response to magnetic ordering at T_C and a magnetic response to the crystallographic phase change at T_S , representing the first demonstration of magnetoelastic coupling in chromium trihalides.

Our results highlight the interesting structural and physical behaviors of CrI_3 and reveal the coupling between them. The detailed characterization, in particular the crystallographic properties, enable future computational studies to further develop an understanding of these materials. The availability of relatively large and easily cleavable single crystals and the presence of coupling between magnetic and structural properties indicate that this understudied class of magnetic vdW bonded insulators presents an attractive target for spintronic-related studies on monolayer and few-layer systems.

RESULTS AND DISCUSSION

Crystal Growth and Room Temperature Structure.

The crystals used in this study were grown by chemical vapor transport, by reacting iodine with chromium powder in a temperature gradient. Further details can be found in the Methods section. The crystals formed as thin and flexible

platelets, with irregular shapes and lateral dimensions up to about 1 cm. The insulating nature of the material was confirmed by measurements of resistivity at room temperature, which gave values of approximately $9 \Omega\text{m}$. As has been noted in the literature, CrI_3 crystals are relatively stable in air, but their decomposition is rapidly accelerated by the presence of CrI_2 , which reacts quickly with moisture.¹⁹ The stability of the CrI_3 crystals is demonstrated in Figure 1a, which shows diffraction

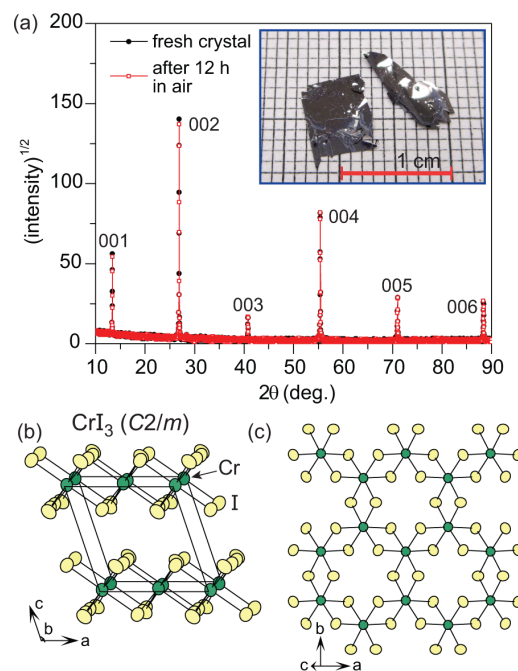


Figure 1. (a) X-ray diffraction patterns from the cleavage plane of a CrI_3 crystal. The platelike habit of the crystals grown by vapor transport is shown in the inset. The two diffraction patterns were collected before and after a 12 h exposure to air. (b) Room temperature crystal structure of CrI_3 , viewed along the monoclinic b axis. (c) Single CrI_3 layer with a viewing direction normal to the ab plane.

patterns from the surface of a crystals collected before and after a 12 h exposure to air. Some of the platelike crystals are shown in the inset of Figure 1a.

Single-crystal X-ray diffraction data collected at 250 K indicated that CrI_3 is monoclinic at this temperature, in contrast to the original structure report.⁸ A starting structural model based on the reported structure of CrCl_3 was used to successfully refine the crystal structure. Refinement results are shown in Table 1, and the structure is shown in Figure 1b,c. The monoclinic structure determined here shows that CrI_3 adopts the AlCl_3 structure type at room temperature. Thus, at room temperature the iodide is isostructural with CrCl_3 but not CrBr_3 , which adopts this structure only above about 420 K.⁹

Crystallographic Phase Transition. To understand how the crystal structure of this material varies with temperature, and in particular to identify any structural phase transitions and to determine whether any structural anomalies occur at the magnetic ordering transition, X-ray diffraction from the surface of as-grown crystals was measured between 300 and 20 K. The results are summarized in Figure 2. The monoclinic 004

Table 1. Crystallographic Information, Selected Interatomic Distances, and Layer Spacings for the High-Temperature Monoclinic and Low-Temperature Rhombohedral Structures of CrI₃

<i>T</i> (K)	250	90
space group	<i>C2/m</i>	<i>R</i> $\bar{3}$ ^a
<i>a</i> (Å)	6.866(3)	6.867(4)
<i>b</i> (Å)	11.886(5)	6.867(4)
<i>c</i> (Å)	6.984(3)	19.807(17)
β (deg)	108.507(7)	90
vol (Å ³)	540.5(4)	809(1)
Cr1 position	0, 0.16654(5), 0	$\frac{1}{3}$, $\frac{2}{3}$, 0.33299(7)
I1 position	0.22407(5), 0, 0.23862(4)	0.31677(9), 0.33453(5), 0.41230(3)
I2 position	0.25110(3) 0.32452(2) 0.23467(4)	
R1/wR2, all data	0.020/0.049	0.057 [0.031] ^b /0.123 [0.069] ^b
Cr–I distances (Å)	2.7267(8), 2.7185(8), 2.7196(8)	2.725(2), 2.727(2)
Cr–Cr distances (Å)	3.959(2), 3.965(1)	3.965(2), 3.965(3)
I–I intralayer short distances (Å)	3.852(1), 3.862(1), 3.863(1)	3.857(2), 3.878(2)
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vdW gap (Å)	3.488	3.471

^aReverse-obverse twinning present with a twin fraction of 0.60.^bValues in square brackets are for the model including a small amount of disorder described in the text and the SI.

reflection from a CrI₃ crystal was monitored as the temperature was varied (Figure 2a). Data were collected in 5 K increments with an average temperature sweep rate of 1.2 K/min for each cooling/warming cycle. This measurement revealed a doubling of this reflection beginning near 180 K and a shift in the intensity between the two reflections upon cooling to about 115 K. This indicates that a temperature-induced crystallographic phase transition indeed occurs in CrI₃ below room temperature and that the high- and low-temperature phases coexist over a wide temperature range. Upon subsequent warming, the transition back to the high-temperature state occurs at a significantly higher temperature of 220 K. In the next cycle, the transition upon cooling occurs at a higher temperature than that upon the initial cooldown, starting near 200 K. The second warming cycle is similar to the first. All measurements show a finite temperature range over which the two phases coexist. This is most pronounced upon initial cooling, suggesting that the as-grown crystals may exist in a strained state and that the strain is released upon cycling through the phase transition. The results of similar measurements on another CrI₃ crystal are shown in Figure 2b. Here the data are plotted as layer spacing (*d*) versus temperature. The data shown were collected after cycling this crystal once through the phase transition, and the coexistence region is highlighted on the plot. The coexistence of the high- and low-temperature phases at intermediate temperature and the thermal hysteresis observed upon cooling and warming demonstrate that the transition is first-order.

A small but clear anomaly in the layer spacing is observed near the Curie temperature of 61 K, which was determined from magnetization and heat capacity measurements discussed

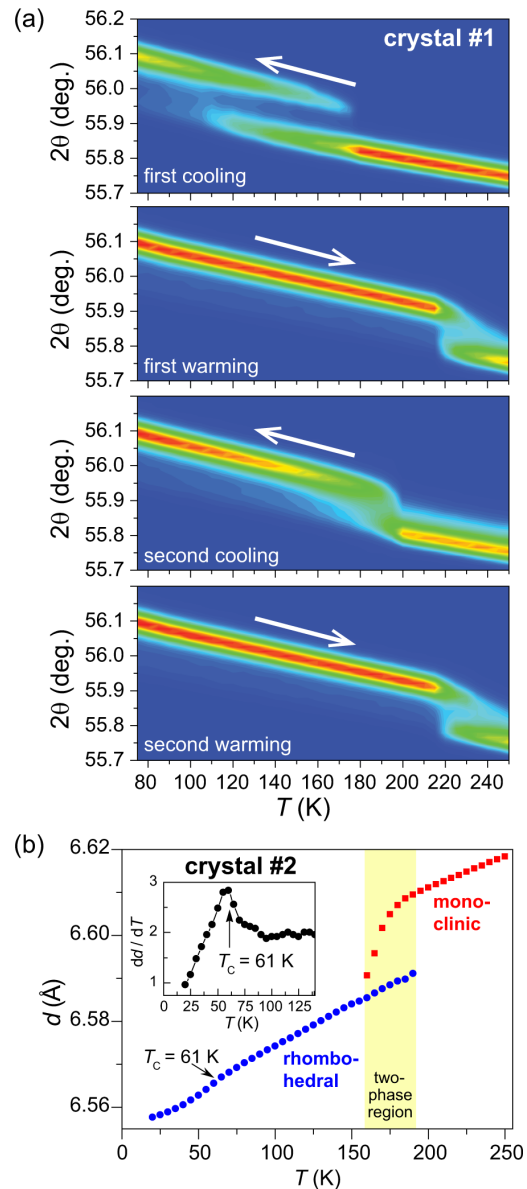


Figure 2. (a) Contour plots of intensity versus diffraction angle and temperature showing the monoclinic 004 reflection measured from a CrI₃ single crystal on two subsequent cooling–warming cycles. (b) Temperature dependence of the spacing (*d*) between CrI₃ layers determined from X-ray diffraction measurements measured upon cooling after cycling through the phase transition once. The effects of magnetoelastic coupling near the Curie temperature are emphasized in the inset, where the temperature derivative is plotted. Different crystals were used to collect the data shown in parts a and b.

below. The nearly linear decrease in *d* upon cooling is seen to accelerate near this temperature, resulting in a sharp cusp in its temperature derivative near *T*_C (Figure 2a, inset). This was observed in every crystal and on every thermal cycle measured in this study. Near the Curie temperature, the ferromagnetic state shows enhanced thermal expansion along the stacking direction, suggesting that the attraction between CrI₃ layers is

affected by the magnetic state of the Cr sublattice. The lattice response at T_C will be addressed further in the discussion of the magnetic properties below.

Low-Temperature Structure. To examine the crystal structure below the transition identified in the data shown in Figure 2, single-crystal X-ray diffraction measurements were conducted at a temperature of 90 K. Several data sets were collected. The best refinement results were obtained from a crystal that had been cooled and warmed through the transition prior to mounting in the cold stream for measurement. This was done to increase the transition temperature, as demonstrated above in the discussion of Figure 2b, and ensure that the entire crystal was transformed to the low-temperature state at the data collection temperature. The results of this structure refinement are shown in Table 1 and discussed below.

Indexing the diffraction data collected at 90 K suggested hexagonal or trigonal symmetry. The data were integrated using low hexagonal symmetry, and primitive space groups were suggested by *XPREP*. However, no reasonable structure solution could be obtained using these space groups. A solution obtained using the space group previously reported for CrI_3 ($P3_112$) has the expected layered structure but very large *R* values ($R1 = 0.39$). Rhombohedral centering was then tried because $R\bar{3}$ structures have been previously reported for CrCl_3 at low temperature and CrBr_3 . Two of the atomic positions determined by direct methods were identified by inspection of the interatomic distances as Cr and I (referred to here as Cr1 and I1), giving $R1 = 0.097$ and $wR2 = 0.254$, and a structure that is like that reported for the rhombohedral chromium trihalides noted above, the BiI_3 structure type. Because rhombohedral symmetry was not detected by the initial analysis of the intensity statistics, reverse-obverse twinning was deemed likely. Reverse-obverse twinning can be considered as a type of stacking fault in this layered structure. Stacking faults are expected to occur at the first-order crystallographic phase transition upon cooling from the monoclinic, high-temperature state, which amounts essentially to a change in the relative translation between adjacent layers in the stacking sequence (see below). The twin law $(0-1\ 0, -1\ 0\ 0, 0\ 0-1)$, a 2-fold rotation about $a - b$,²³ was included in the model, and the agreement indices were reduced to $R1 = 0.057$ and $wR2 = 0.123$. This indicates that the basic structural model, the BiI_3 type, is the correct description for the structure of CrI_3 at low temperature.

The remaining discrepancy between this structural model and the diffraction data is attributed to further stacking disorder developed during the monoclinic to rhombohedral phase transition, beyond that accounted for by the reverse-obverse twinning. The strongest difference peaks were identified as partially occupied Cr and I sites to construct the partially disordered model. The disorder is not strong; the occupancies of the partial sites refined to about 4%. The resulting disordered structure model gives $R1 = 0.031$ and $wR2 = 0.069$ (all data) and is shown and discussed in more detail in the Supporting Information (SI).

It is reasonable to assume that the disorder and twinning used to describe the low-temperature diffraction data represent extrinsic effects and that the BiI_3 type describes the ideal structure of CrI_3 below about 200 K. The CrI_3 layers are essentially identical in both the high- and low-temperature structures. Bond valence sum calculations for Cr in the high- and low-temperature structures give 2.9 and 2.8, respectively, consistent with the expected 3+ oxidation state of Cr. The

crystallographic features of the monoclinic and rhombohedral structures are compared in Figure 3.

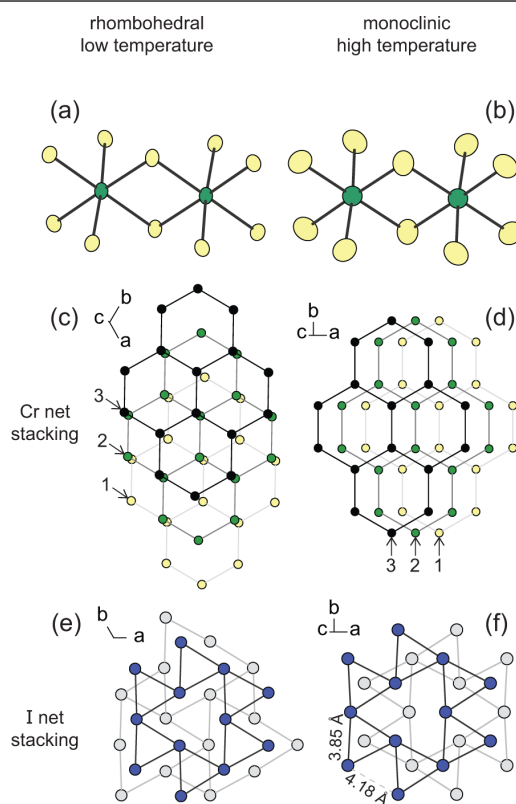


Figure 3. Comparison of the stacking sequences of the Cr and I 2D nets in the high- and low-temperature structures of CrI_3 . The view is slightly offset from the stacking direction for the rhombohedral structure. The two I nets shown for each structure are those separated by the vdW gap.

The coordination environments around two neighboring Cr atoms are shown in Figure 3a,b, with interatomic distances listed in Table 1. The atoms are shown as ellipsoids representing their refined anisotropic atomic displacement parameters. As expected, the I ellipsoids are elongated toward the holes in the Cr honeycomb net, indicating their tendency to vibrate toward these voids. Upon cooling from the $C2/m$ structure at 250 K to the $R\bar{3}$ structure at 90 K, there is little change in the geometry of the CrI_3 slabs, as indicated by the Cr–I and Cr–Cr distances. In terms of interatomic distances, the main crystallographic changes are a decrease in the interlayer spacing (defined as the perpendicular distance between two Cr sheets) and the vdW gap (defined as the perpendicular distance between the I sheets on each side of the gap).

In both structures, the I atoms form distorted close-packed nets (Figure 3e,f) on each side of the honeycomb net of Cr atoms (Figure 3c,d). The Cr atoms cap the smaller triangles in the I net. The crystallographic phase transition involves shearing of the vdW bonded layers to change their relative positions upon stacking perpendicular to the ab plane (Figure 3c–f). In the rhombohedral structure, the Cr nets are displaced upon stacking so that the Cr in one layer is directly over the

center of a hole in the honeycomb net of the two adjacent layers, in an ABC stacking sequence. In the monoclinic structure, each subsequent layer is displaced along the a direction so that the first layer is nearly, but not quite, eclipsed by the fourth layer. Thus, the stacking in the high-temperature structure is nearly ABC, but the displacements between the A, B, and C layers is different from that found in the low-temperature structure. A comparison with the previously reported trigonal structure can be found in the SI.

Interestingly, the collapse in the layer spacing upon cooling through the phase transition noted in the diffraction results in Figure 2 is not clear in the single-crystal data (Table 1). Indexing single-crystal diffraction patterns from several crystals at 90 K indicated layer spacings ranging from 6.58 to 6.60 Å, suggesting that this could perhaps be due to experimental uncertainty or crystal-to-crystal variations. Alternatively, the difference in the strain states developed upon cooling between crystals mounted on a flat plate with N-grease for powder diffractometer measurements and those mounted by immersion in paratone oil for the single-crystal diffractometer measurements may be responsible. The latter would suggest that pressure may provide a way to tune the structural and magnetic properties of this material.

Structural Stability and Interlayer Binding from Density Functional Theory (DFT). The relative stability of the two phases was examined using DFT. To do this accurately, special consideration must be given to the weak, long-ranged forces binding the CrI₃ layers, which are not captured by most exchange-correlation functionals like the local density approximation or the generalized gradient approximation (GGA). Here, the van der Waals density functional (vdW-DF), which has been developed specifically to incorporate these dispersion forces,^{24,25} is used. The optB86b exchange functional was employed.²⁶ A comparison of the computed formation energies of the two phases indicated the rhombohedral phase of CrI₃ to be more stable at 0 K but only by about 2 meV/atom. Both phases were found to have negative formation enthalpies (Table 2). The energy difference between the two phases is close to the expected accuracy limit for these types of calculations;²⁷ however, the findings are consistent with the

Table 2. Crystallographic Properties from Structural Optimizations and Formation Energies (E_{form}) Determined Using DFT Calculations

	vdW-DF-optB86b	GGA
	$R\bar{3}$	
a (Å)	6.889	7.10
c (Å)	19.797	21.81
vol (Å ³)	814	952
layer spacing (Å)	6.599	7.27
Cr–I (Å)	2.96	2.73
E_{form} (eV/atom)	−0.180	
	$C2/m$	
a (Å)	6.843	7.030
b (Å)	11.845	12.20
c (Å)	6.976	7.314
β (deg)	108.96	107.6
vol (Å ³)	534	598
layer spacing (Å)	6.240	6.972
Cr–I (avg) (Å)	2.70	2.73
E_{form} (eV/atom)	−0.178	−

experimental results presented here. Their closeness in energy, and the entropic contributions to the Gibbs free energies at finite temperatures, likely results in the observed phase transition between the rhombohedral and monoclinic structures. Calculations of interlayer binding energy curves (shown in the SI) indicate that the difference in the formation energies is accounted for by the difference in vdW bonding between the CrI₃ layers, consistent with the very similar intralayer geometries noted in crystallographic analysis. Thus, this energy difference, although small, is consistent with a structural transition driven by optimization of the vdW bonding between the CrI₃ layers.

Table 2 shows the calculated crystallographic properties obtained by allowing the structural parameters (lattice constants and atomic positions) to relax while minimizing the total energy. Results obtained using vdW-DF-optB86b are compared with experimental numbers and with results obtained using GGA, which lacks vdW interactions. This shows that vdW-DF-optB86b does a remarkably good job of capturing the interatomic interactions in this material, while GGA, which does not include the longer ranged forces, significantly overestimates the layer spacing. This demonstrates the importance of properly modeling the vdW forces in computational studies of the structural and electronic properties of these trihalides, as well as other 2D-layered materials that are of interest for electronic and spintronic applications.

From the perspective of potential electronics applications, a key property of this material is its cleavability. The crystals are observed to be easily cleaved, and calculations of the interlayer binding energies support this. For the calculations, a structural model was constructed that contained three layers of CrI₃ separated from one another by the equilibrium distance (d_0), with a variable distance (d) between each successive three-layer block as they are stacked along the c axis (Figure 4).

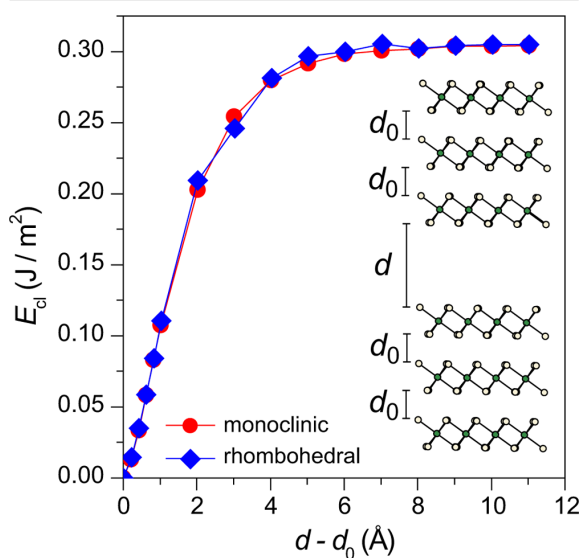


Figure 4. Calculated energy (E_d) required to cleave CrI₃ into three-layer thick slabs separated by a distance d relative to the equilibrium separation d_0 . The results for both the monoclinic and rhombohedral structures are shown, and in both cases, E_d approaches 0.30 J/m² at large separation. Calculations were performed using vdW-DF-optB86b.

Calculations were performed for both the monoclinic and rhombohedral stacking sequences, and in-plane structures were fully relaxed at each point. The cleavage energy was calculated by determining the total energy difference (per unit area) of the fractured structure as a function of d , with respect to the equilibrium ground state. The results are shown in Figure 4. For large separations, the cleavage energy converges to a value of 0.30 J/m^2 for both phases. Similar calculations using single layers instead of the trilayers shown in Figure 4 gave results within a few percent of those shown here. This energy can be compared to the cleavage energy of other vdW-bonded materials of interest. Similar calculations have given cleavage energies of 0.35 and 0.38 J/m^2 for CrSiTe_3 and CrGeTe_3 , respectively,¹² while 0.43 J/m^2 has been calculated for graphite and 0.27 J/m^2 for MoS_2 .²⁸

Magnetic Properties: Experiment and Theory. As noted in the Introduction, there are few reports of magnetic properties of CrI_3 . In addition, the published data are limited to measurements on polycrystalline material and therefore contain little information regarding magnetic anisotropy. Sparse magnetic susceptibility versus temperature data from CrI_3 powder were reported by Hansen and Griffel,²⁰ but the Curie temperature is not apparent in the data. Hansen also reported magnetization versus applied field for CrI_3 powder measured at 4.2 K .¹⁸ The data demonstrate the ferromagnetic nature of the compound at this temperature, but the applied fields were too low to saturate the Cr magnetic moment. Finally, Dillon and Olson describe the results of more detailed magnetization measurements including the identification of the Curie temperature (68 K), saturation moment ($3.10 \mu_{\text{B}}/\text{Cr}$), effective paramagnetic moment ($4.00 \mu_{\text{B}}/\text{Cr}$), and Weiss temperature (70 K), although no data are shown in the paper.¹⁷ No previous studies have considered the possible effects of the crystallographic phase transition on the magnetic behavior of this or other chromium trihalides.

The results of our magnetization measurements on CrI_3 crystals are shown in Figure 5. Measurements were performed with the field both parallel and perpendicular to the c^* axis, which is the direction along which the CrI_3 layers are stacked (the c direction in the hexagonal setting of the rhombohedral structure), giving insight into the magnetic anisotropy of the material. Temperature-dependent magnetization data, collected upon cooling in applied fields ranging from 100 Oe to 50 kOe , are shown in Figure 5a. The data clearly show the ferromagnetic nature of the crystals, with a Curie temperature near 61 K determined from the lowest field data shown in the inset. This is supported by the magnetization versus applied field data shown in Figure 5b. Below the Curie temperature, the magnetization measured along the c axis saturates at a relatively low magnetic field, indicating that the moments are aligned in this direction. The saturation moment at the lowest temperature is $3.1 \mu_{\text{B}}/\text{Cr}$, close to the expected value of $gS = 3 \mu_{\text{B}}$ expected for trivalent Cr with electronic configuration $3d^3$. The ferromagnetically aligned moments in the layered Cr compounds CrSiTe_3 and CrGeTe_3 , which have Curie temperatures of 33 and 61 K , respectively, are also directed normal to the crystallographic layers, with reported saturation moments ranging from 2.7 to $3.0 \mu_{\text{B}}/\text{Cr}$.^{29–31}

The magnetic anisotropy can be seen by a comparison of the data in parts a and b of Figure 5 for the two orientations of the magnetic field. In the temperature dependence shown in Figure 5a, the magnetization is nearly isotropic at 50 kOe , while at lower fields, significant magnetic anisotropy is observed below

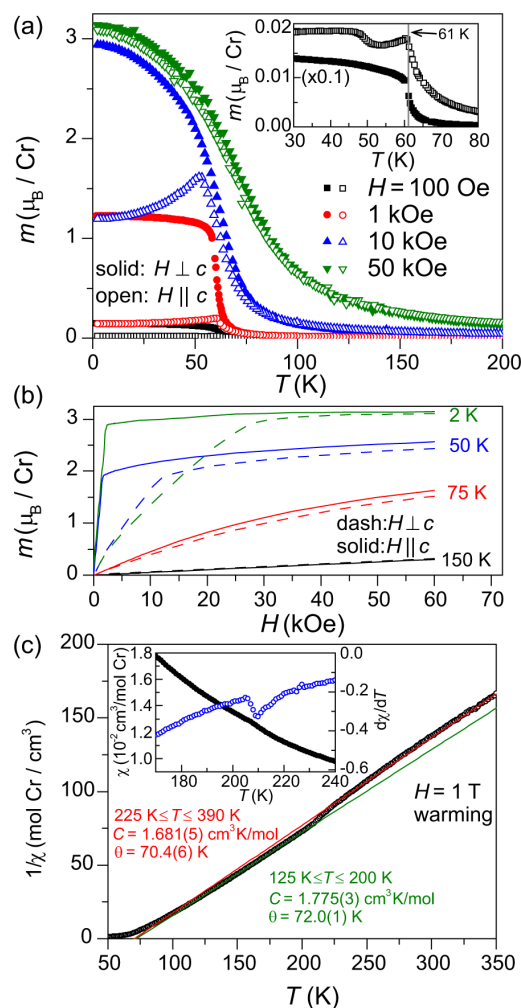


Figure 5. Anisotropic magnetic properties of CrI_3 single crystals. (a) Magnetic moment versus temperature measured at the indicated magnetic fields with the field applied parallel and perpendicular to the stacking direction (c^* axis). (b) Magnetic moment versus applied magnetic field measured in both orientations. (c) Magnetic susceptibility of a collection of many CrI_3 crystals measured upon warming through the rhombohedral to monoclinic phase transition, illustrating the magnetic anomaly at T_S .

T_C . This is also seen in the m versus H data collected at 2 and 50 K shown in Figure 5b. The data show that the ordered magnetic moments lie along the c axis, normal to the ab plane in which the CrI_3 layers lie because that is the direction along which the magnetization is most easily saturated. This is consistent with reports from Mössbauer spectroscopy.^{21,22} When the field is applied along the easy axis (normal to the CrI_3 layers), the magnetization rises sharply to the saturation value. When the field is applied perpendicular to this axis, the moments gradually and smoothly rotate away from the easy axis, toward the direction of the applied field. This is behavior typical of anisotropic ferromagnets.³² The anisotropy can be quantified by defining the anisotropy field, which is approximately the field at which the m versus H curve measured perpendicular to the easy axis saturates. Figure 5b

shows that this is near 30 kOe at 2 K, consistent with the value deduced by ferromagnetic resonance techniques,¹⁷ and decreases as the temperature is raised toward T_C . A consequence of this is the divergence of the m versus T curves measured in the two directions at 10 kOe and below (Figure 5a). Full magnetization loops are not shown because the remanent magnetization was observed to be small (at most 3% of the saturation magnetization) and little coercivity was seen (coercive fields of less than 100 Oe). This soft ferromagnetism occurs despite the large magnetic anisotropy in CrI_3 because of the single-crystal nature of the measurement sample. In large single crystals, the magnetization can be easily removed as the field is reduced to zero by the formation of magnetic domain walls that move with little resistance through the crystal lattice.³² Similar magnetization behavior has been reported for crystals of CrBr_3 ¹⁶ and CrGeTe_3 ,³¹ although both of these materials have significantly smaller magnetic anisotropy than CrI_3 . Note that the low-field data measured perpendicular to the easy axis show an unusual temperature dependence below T_C . This may either be a consequence of the temperature-dependent magnetic anisotropy or perhaps indicate magnetic order that is more complex than simple ferromagnetism.

In the extreme, spintronic applications would use magnetic materials in the form of single monolayers. Therefore, it is interesting and important to consider how the magnetic properties of layered materials might change as a function of the thickness. Here, first-principles calculations, as described in the Methods section, were used to examine the magnetic ground state of CrI_3 in bulk and monolayer forms. Calculated electronic band structures and densities of states (DOSs) for the high- and low-temperature bulk structures are shown in the SI. As expected, nonmagnetic calculations predict metallic behavior, with the Fermi level in the middle of the half-filled $\text{Cr } t_{2g}$ bands. Calculations show that the spin-polarized models are lower in energy than the nonmagnetic models by about 1.7 eV per formula unit, indicating a strong tendency toward magnetism. The ferromagnetic model is lower in energy than the antiferromagnetic model by about 20 meV per formula unit, consistent with the experimentally observed ferromagnetic order. Magnetic moments of $3.0 \mu_B/\text{Cr}$ are calculated, in good agreement with the saturation moment measured experimentally. Magnetic ordering opens a gap at the Fermi level in the calculated DOSs. Experimentally this material is insulating even at room temperature in the paramagnetic state. Magnetic order cannot be responsible for the gap at room temperature, which suggests that CrI_3 is a Mott insulator.

To examine the monolayer system, a structure in which the individual CrI_3 layers are separated by large distances was simulated and the energies of magnetically ordered states were compared to that determined by nonmagnetic calculations. Calculations were performed on monolayers extracted from both the rhombohedral and monoclinic structures of CrI_3 . For this purpose, GGA was used; however, because of the large separation between the layers used in the calculation, the effects of not treating the vdW forces properly are expected to be minimal. Indeed these calculations show the ferromagnetically ordered state to be energetically favored over the nonmagnetic state. The calculated ordered moment is $3.0 \mu_B/\text{Cr}$, and the ferromagnetic state is found to be 1.5 eV lower in energy (per formula unit) than the nonmagnetic state. These are similar to the results obtained from GGA calculations on the bulk structures in both the rhombohedral and monoclinic structures noted above and to the results for bulk chromium halides

reported in ref 11. These preliminary calculations suggest that the ferromagnetism found in bulk CrI_3 crystals may persist into the monolayer state, as has been predicted for the layered magnetic material CrSiTe_3 ,¹² which is important when considering a material's potential for applications. Monte Carlo simulations, which include the effects of magnetic fluctuations, could provide an estimate of the Curie temperature of the monolayers.

As noted above, a structural response is observed in the layer spacing when long-range magnetic order sets in at T_C , indicating coupling between structure and magnetism. The nature of this coupling could be elucidated by a theoretical study that combines both magnetism and structural optimization treating the vdW interactions between the layers accurately. This is beyond the scope of the present work but is expected to be addressed in a forthcoming study. Because long-ranged magnetic order is observed below a relatively high Curie temperature ($T_C = 61$ K) in CrI_3 , there must exist relatively strong magnetic exchange interactions across the vdW gaps, suggesting that magnetism should contribute significantly to the energetics associated with stacking of the layers. Because the chemical bonding between the layers is weak, it may be expected that the magnetic attraction between neighboring layers in the ferromagnetic state could be the driving force for contraction along the stacking direction when cooling through T_C . The data in Figure 2b clearly show that the lattice contracts along the stacking direction as the ferromagnetic state develops, consistent with this intuitive argument. Although the magnetic interactions are expected to be strongest within the CrI_3 layers, the intralayer chemical bonding is much stronger than the vdW bonds between the layers, so the in-plane magnetoelastic response, which is not probed in this work, may be much weaker. A change in the crystallographic symmetry may also occur at this temperature, and single-crystal diffraction at very low temperatures would be required to examine this possibility. It is interesting to note that the reentrance of a high-temperature crystal structure has been observed at the Curie temperature in the layered ferromagnet $\text{BaFe}_2(\text{PO}_4)_2$.³³

To demonstrate the coupling between the structure and magnetism in CrI_3 further, the magnetic behavior was carefully examined near the crystallographic phase transition at T_S . Indeed, an anomaly in the magnetic susceptibility is observed. To clearly distinguish this feature, measurements were performed on a collection of nonoriented crystals having a significantly larger mass than the coaligned single crystals used for the measurements discussed above. The resulting powder-averaged magnetic susceptibility ($\chi = M/H$) versus T data are shown in Figure 5c. The data were measured upon warming because the crystallographic studies discussed above showed the sharpest transition when warming. The main panel of Figure 5c shows the inverse susceptibility from 50 to 350 K. A subtle feature is indeed observed at 210 K in χ versus T , as shown in the inset. This is more clearly evident in the temperature derivative, which is also shown. The $1/\chi$ data is seen to be linear over a relatively wide range of temperatures both above and below T_S . Fits to the Curie–Weiss law using these two temperature ranges are shown on the figure. The Weiss temperatures (θ) are positive and near 70 K, consistent with the observed ferromagnetic ordering near 61 K. The effective moment determined from the Curie constants (C) are 3.67 and $3.77 \mu_B/\text{Cr}$ above and below T_S , respectively, displaying an appreciable difference. A similar value, $3.77 \mu_B/\text{Cr}$, has been reported for CrGeTe_3 .³⁰ The expected spin-only

value is $3.87 \mu_B/\text{Cr}^{3+}$. This subtle but easily detectable difference in the magnetic behavior of the monoclinic and rhombohedral states is a clear manifestation of the coupling of the crystal structure to the magnetism in CrI_3 .

The heat capacity of CrI_3 crystals is shown in Figure 6. The data approach the Dulong–Petit limit at high temperatures, and

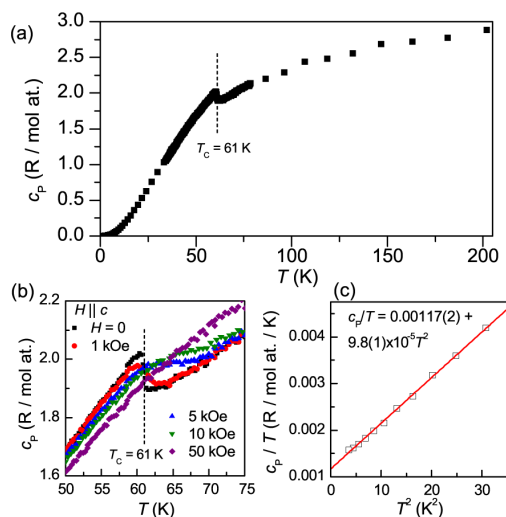


Figure 6. Specific heat capacity of CrI_3 single crystals. (a) Sharp anomaly observed at the Curie temperature in zero magnetic field. (b) Effect of an applied magnetic field on this anomaly, which is consistent with expectations for a ferromagnet. (c) Low-temperature data plotted as c_p/T versus T^2 , giving a slope related to the phonon heat capacity and an intercept related to the magnetic contribution.

a clear anomaly is observed at T_C . Efforts to observe any anomalous behavior near the structural phase transition were not successful. Figure 6b shows the effect of an applied magnetic field on the lambda-like feature at T_C . The response is typical of a ferromagnetic transition, with a broadening toward higher temperatures observed as the magnetic field is increased.³⁴ The entropy associated with the peak is small, about $0.14R/\text{mol}$ of Cr . Analysis of CrBr_3 heat capacity data by Jennings and Hansen suggests that a relatively small amount of the total magnetic entropy $[=R \ln(2S + 1)]$ is associated with the peak at the Curie temperature due to short-ranged or 2D magnetic correlations existing at higher temperatures, up to about $4T_C$.³⁵ The Curie–Weiss behavior noted above suggests that in CrI_3 magnetic correlations are probably not very strong above about 125 K, approximately $2T_C$. Further analysis would require more information about spin–spin correlations in this material. Inspection of the low-temperature behavior in the CrI_3 data (Figure 6c) shows a T linear term in addition to the expected T^3 phonon contribution. Because CrI_3 is an insulator, this linear term is not attributed to the usual electronic heat capacity but rather to magnetic excitations associated with the 2D ferromagnetic nature of this material.³⁶ From the slope, β , of the linear fit in Figure 6c, the Debye temperature, Θ_D , can be estimated from $\beta = 234/\Theta_D^3$ in the units used in the figure.³⁷ This gives a Debye temperature of 134 K.

CONCLUSIONS

This paper reports the detailed crystallographic and magnetic properties of CrI_3 at and below room temperature and presents

evidence for coupling between the magnetic and structural behaviors. A first-order structural phase transition from the monoclinic AlCl_3 structure type at high temperatures to the rhombohedral BiI_3 structure type below about 200 K is identified. DFT results suggest that optimization of the vdW bonding between the CrI_3 layers may be the driving force for the transition to the rhombohedral structure upon cooling. This crystallographic transition is manifested in the magnetic behavior as an apparent increase in the effective magnetic moment of Cr. Strong magnetic anisotropy is observed below the Curie temperature of 61 K, with ordered magnetic moments directed perpendicular to the CrI_3 layers. A lattice response is observed at this magnetic phase transition, with enhanced thermal expansion below T_C , suggesting that the long-range magnetic order influences the interlayer bonding significantly. First-principles calculations employing the vdW-DF accurately reproduce the experimental structures and correctly determine their relative stability. Calculated cleavage energies are comparable with those of other layered materials of current interest [graphite, MoS_2 , and $\text{Cr}(\text{Si}/\text{Ge})\text{Te}_3$]. In addition, calculations identify the preference for ferromagnetic order with an ordered moment of $3 \mu_B/\text{Cr}$ and suggest that this magnetic ordering may persist in thin samples down to monolayer thickness. The present experimental and theoretical understanding of CrI_3 suggests that it will be of great interest for study in single- or few-layer forms and identifies layered transition-metal halides, and the chromium trihalides in particular, as an attractive class of materials with potential impact on post silicon electronics research, which aims to include magnetism in device functionality.

METHODS

The crystals of CrI_3 were obtained by reacting chromium powder (99.999%) with anhydrous iodine beads (99.999%) in a 1:3 molar ratio inside evacuated and sealed silica tubes of length 15 cm, inner diameter 22 mm, and wall thickness 1.5 mm. The tubes were placed in the natural temperature gradient present in an open-ended tube furnace, with the charge (total mass ranging from 0.4 to 0.8 g) at one end of the tube near the center of the furnace. The furnace temperature was raised to 650 °C and held there for several days. The crystals grew near the cooler end of the tube, where the temperature ranged from approximately 500 to 550 °C. The crystal growth proceeded rapidly, with visible crystals forming after only a few hours at temperature; however, relatively little of the starting material was transported even after 6 days, suggesting that the growth slows considerably as the iodine is consumed. No crystals formed at hot-end temperatures of 600 and 500 °C. Clean single crystals are stable in laboratory air for at least 1 day, but to avoid degradation due to CrI_2 contamination,¹⁹ the samples were kept under vacuum when stored for extended periods of time. The crystals are thin, shiny, and black and delaminate easily.

X-ray diffraction from the faces of single crystals was measured with a PANalytical X'Pert Pro MPD diffractometer, with monochromatic $\text{Cu K}\alpha_1$ radiation. The samples were placed on a flat aluminum sample holder using a light film of cryogenic grease (Apiezon N-grease). Measurements were performed in vacuum, and the temperature was controlled with an Oxford Phenix closed-cycle helium cryostat. Full single-crystal diffraction data sets were collected using a Bruker APEX CCD diffractometer using $\text{Mo K}\alpha$ radiation. Samples were mounted in a film of Paratone oil suspended within a nylon loop to minimize bending of the thin crystals, which were approximately $130 \times 70 \times 10 \mu\text{m}^3$. Collection temperatures, 250 and 90 K, were controlled with a nitrogen cold stream. The crystal used for the low-temperature data collection reported in Table 1 was first cycled from room temperature to 5 K and back before mounting. Data collection and indexing were performed with Bruker SMART software. Reflections were harvested

and data integrated using SAINTPlus. Empirical absorption corrections were made with SADABS, and symmetry analysis and further data preparation were performed with XPREF. Structure solution and refinement were carried out with the ShelXS-2013 and ShelXL-2013³⁸ program suite, respectively, within WinGX.³⁹

Magnetization measurements were performed with a direct-current SQUID magnetometer (Quantum Design, Magnetic Property Measurement System). To obtain the anisotropic data shown in Figure 5a,b, several crystals were stacked together with a common [001] direction with a total mass of 0.38 mg. Measurements were then performed with the field parallel and perpendicular to this direction. To reach the precision required to observe with confidence the subtle feature in magnetization near T_S (Figure 5c), 12.9 mg of nonoriented crystals was enclosed in a gelcap. Heat capacity measurements were carried out on a stack of CrI₃ crystals, with a total mass of 1.69 mg, using a Physical Property Measurement System from Quantum Design. Standard, relaxation-type measurements were conducted over the entire temperature range studied, and subsequent measurements around T_C were performed using the “dual-slope” analysis of the heating and cooling curves associated with relatively large heat pulses (temperature rise ~ 10 K). The magnetic field was applied along the [001] direction. The in-plane resistivity was measured at room temperature by contacting CrI₃ crystals with indium-tipped, spring-loaded, gold pins.

Theoretical investigations were performed using DFT with the projector augmented plane-wave (PAW) method,^{40,41} as implemented in the VASP 5.3.3 package.^{42,43} The PAW potentials used explicitly treat 12 valence electrons for Cr ($2p^6 3d^5 4s^1$) and 7 for I ($5s^2 5p^5$). An energy cutoff of 550 eV was used with Monkhorst–Pack special k -point grids of $4 \times 4 \times 2$ and $4 \times 2 \times 4$ for the low-temperature $R\bar{3}$ and high-temperature $C2/m$ phases, respectively. Ions were relaxed until the forces on each atomic site were below 10 meV/Å, simultaneously achieving a total energy convergence of 10^{-6} eV. The interlayer binding energies are calculated using the vdW-DF^{24,25} for correlation and the optB86b functional for exchange.²⁶ The vdW-DF-optimized structures are used to calculate the formation energies of the $R\bar{3}$ and $C2/m$ phases. Further, the electronic band structures, shown in the SI, were computed on top of the vdW-DF-optimized structures using the Perdew, Burke, and Ernzerhof (PBE-GGA)⁴⁴ exchange-correlation functional.

■ ASSOCIATED CONTENT

Supporting Information

Description of the disorder model used for the low-temperature structural refinements, comparison of structures determined here with a previously reported structural model in space group $P3_12$, and electronic band structures, DOSs, and binding energy curves from density functional theory calculations. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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